

Questions

Question 1:

We had you define the acceleration due to gravity as a field in a structure that you had to pass as an input argument to several functions. Instead, we could have had you type the value for the constant, 3.72 m/s², directly in those functions. Do you believe there is an advantage to how we had you do it? Explain. Would you have done it differently? Explain why or why not.

Placing the gravitational acceleration in a dictionary makes it easy to edit if the value needs to be changed, without having to find and update every script/function where it's used. Having the gravitational acceleration value within the dictionary makes it easier to call when coding as well.

Question 2:

What happens if you try to call $F_{gravity}$ using a terrain slope of 115 degrees? Is this desirable behavior? Explain why you think this.

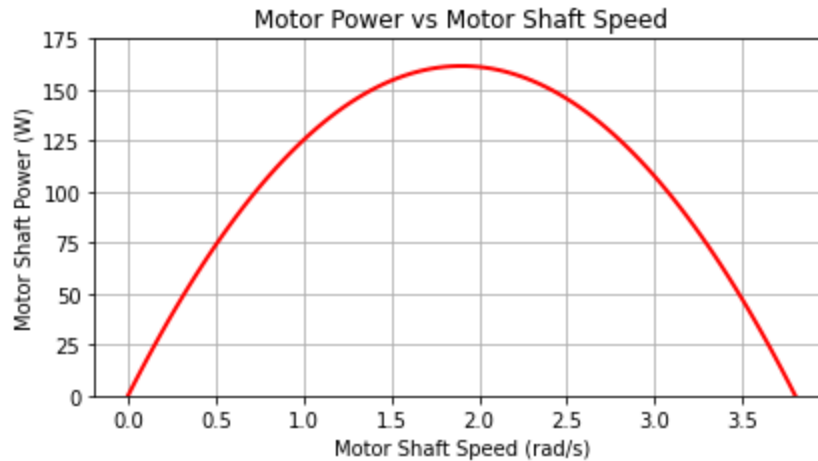
Calling $F_{gravity}$ with a terrain slope of 115 degrees results in an Exception being raised since the angle should be between -75 and +75 degrees. I believe in this scenario, this is desirable because that angle is the critical angle. The critical angle is the angle before the rover would slip and fail. The mentioned 115 degrees would have resulted in the rover failing since it is much higher than the critical angle.

From a physical perspective 115 degrees would be an overhang. The math relies on the force of gravity holding the rover onto the ground. However on an overhang, this would not be the case as the rover would fall away from the overhang unless external equipment was used.

Question 3:

What is the maximum output by a single rover motor? At what motor shaft speed does this occur?

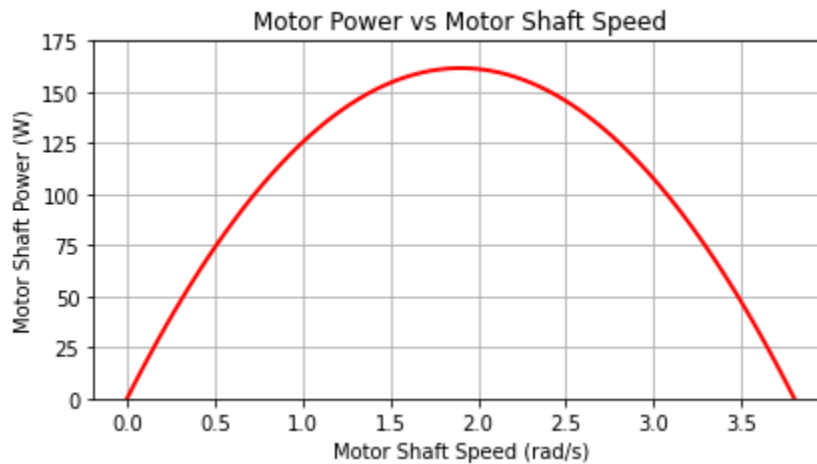
As shown in the below plot, the maximum power output is about 160 Watts. This is represented by the peak of the Motor Power vs Motor Shaft Speed graph. The maximum power output occurs at about 2.0 rad/s which is indicated by the graph which is about half of the maximum shaft speed, 3.8 rad/s.

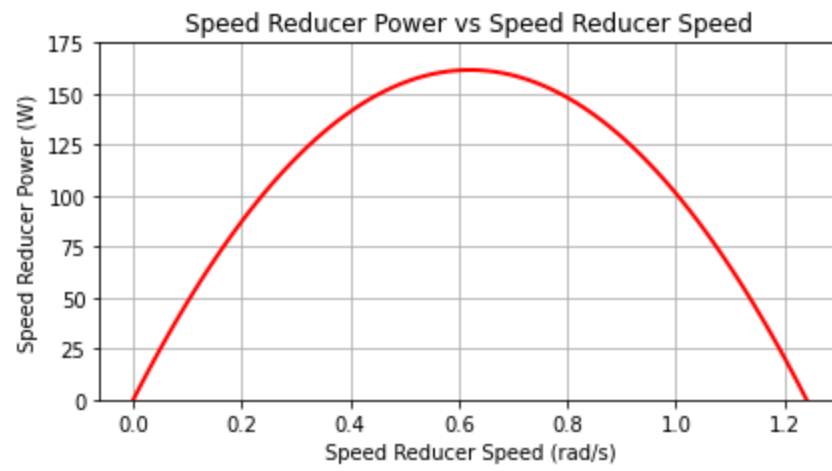


Question 4:

What impact does the speed reducer have on the power output of the drive system?

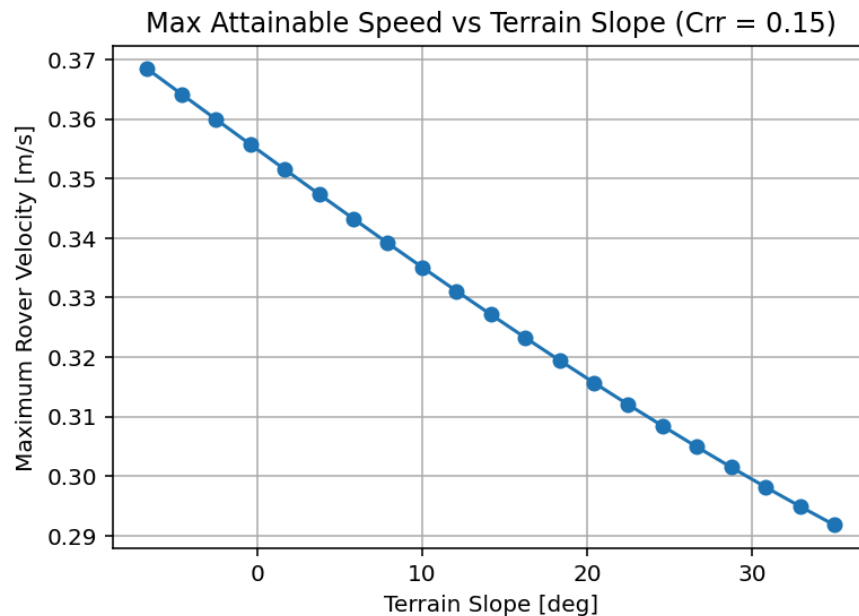
The speed reducer has no effect on the power output of the system. As seen in the graphs below, the power vs shaft speed has the same shape for both reduced and regular speed and appears to reach the same y-values. The speed reducer appears to have been canceled out from torque increasing and shaft speed decreasing by the same magnitude due to the gear ratio.





Question 5:

Examine the graph you generated using `analysis_terrain_slope.py`. (Provide the graph in your response for reference.) Explain the trend you observe. Does it make sense physically? Why or why not? Please be precise. For example, if the graph appears linear or non-linear, can you explain why it should be the way you observed? Refer back to the rover model and how slope impacts rover behavior.



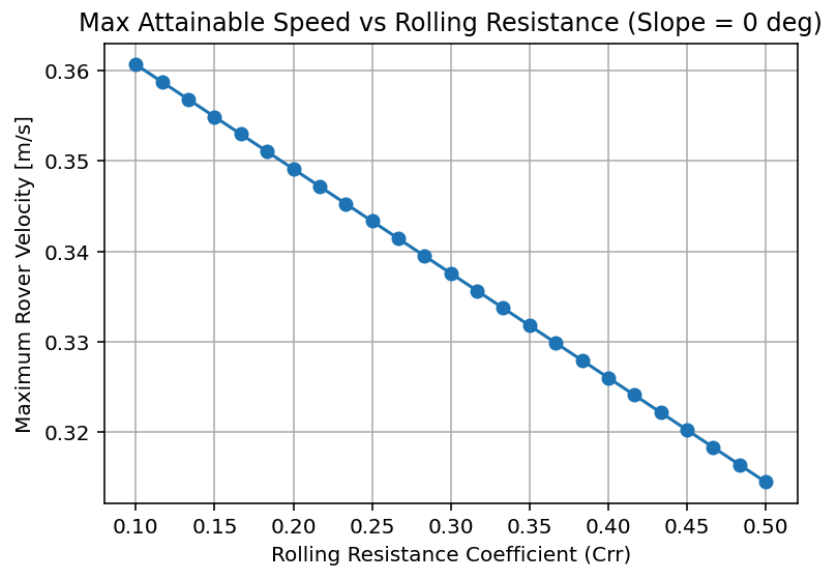
The trend presents as a downward sloping trend. The maximum velocity starts at its highest point when slope = -6.6 degrees and decreases at a constant rate as slope approaches 35 degrees. There are also blank NAN spaces when the slope is less than -8.53 degrees caused by runaway acceleration detailed below.

This trend is perfectly justifiable and expected. As the rover climbs steeper hills, a component of its weight increasingly resists forward motion. As the DC motor has to increase torque to combat this, the top speed drops. However when travelling down hills, a component of weight increasingly helps the rover move forward. This is seen for slight downward slopes. If the angle increases too much, then the acceleration due to a component of weight creates a runaway acceleration.

It is not linear because both force of gravity and force of rolling resistance depend on trigonometric functions. Therefore it is a trigonometric relationship and not an algebraic one.

Question 6:

Examine the graph you generated using `analysis_rolling_resistance.py`. (Provide the graph in your response for reference.) Explain the trend you observe. Does it make sense physically? Why or why not? Please be precise. For example, if the graph appears linear or non-linear, can you explain why it should be the way you observed? Refer back to the rover model and how the coefficient of rolling resistance impacts rover behavior.



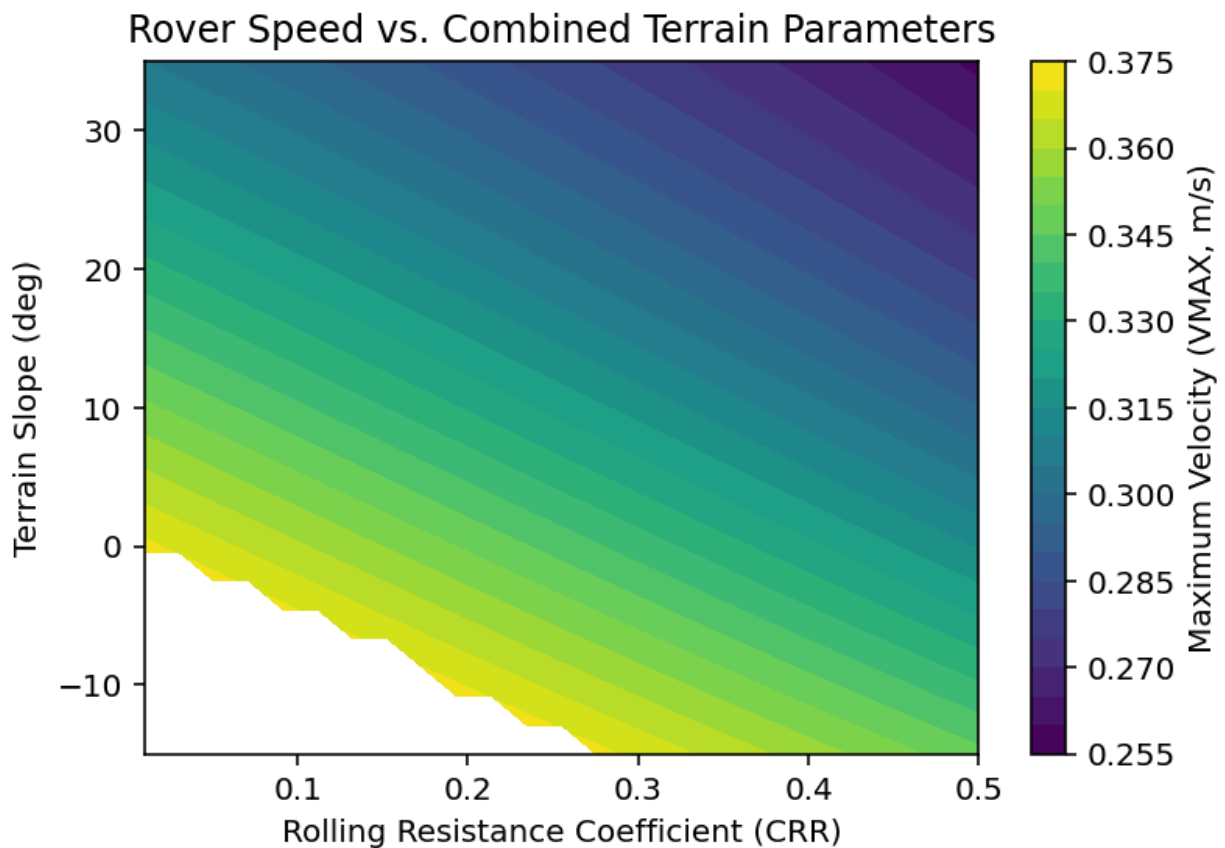
The trend presents as a downward sloping linear line. The maximum velocity starts at its highest point when $C_{rr} = 0.1$ and decreases at a constant rate as C_{rr} approaches 0.5.

This is perfectly justifiable and expected. The coefficient of resistance represents a function of the energy lost due to the terrain. Therefore as C_{rr} increases, the terrain exerts a stronger retarding force against the wheels. To overcome this resistance, the motors must provide more torque thus slowing the DC motors down and lowering the top speed.

It is linear because when perfectly flat (slope = 0), only the force of rolling resistance and the drive force govern the motion. Those are both linear and therefore so is the final result.

Question 7:

Examine the surface plot you generated using `analysis_combined_terrain.py`. (Provide the graph in your response for reference.) What does this graph tell you about the physical conditions under which it is appropriate to operate the rover? Based on what you observe, which factor, terrain slope or coefficient of rolling resistance, is the dominant consideration in how fast the rover can travel? Please explain your reasoning



Based on the results shown in the graph above, both of the terrain conditions (angle and rolling resistance coefficient) significantly contribute to the maximum speed at which the rover can move. For the most efficient operation of the rover, there must be a relatively lower slope and a decreased rolling resistance coefficient (CRR). Furthermore, between these two terrain constraints, the increasing terrain angle seems to be the dominating factor contributing to decreased max velocity. As seen in the figure above, the top-left corner indicates the influence of the terrain slope alone in a teal/turquoise color, whereas the bottom-right corner displays the outcome of CRR alone in a lighter green color. These colors correspond to the maximum velocity as indicated by the colorbar on the left of the graph, which shows teal at a lower maximum velocity than green, proving that terrain slope is the dominant consideration for maximum velocity